

BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

Bluestock Mutual Fund Analytics Platform

Capstone Project Report

Submitted by:

- Sricharan Medaboina
- Team Members

Technologies Used:

- Python
- Pandas
- NumPy
- SQLite
- Power BI
- Jupyter Notebook

Academic Year: 2025- 2026

Executive Summary

The Bluestock Mutual Fund Analytics Platform is a comprehensive data analytics solution developed to analyze mutual fund performance, investor behavior, industry trends, and portfolio risk. The project combines data engineering, exploratory data analysis, performance measurement, advanced risk analytics, and dashboard visualization into a single integrated platform.

The project begins with collecting and cleaning multiple mutual fund datasets.

An ETL pipeline was developed using Python to automate extraction, transformation, and loading processes. Cleaned data was stored in a SQLite database and analyzed using Python libraries such as Pandas and NumPy.

Several performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated. Advanced analytics techniques such as Value at Risk (VaR), Conditional Value at Risk (CVaR),

investor cohort analysis, SIP continuity analysis, and portfolio concentration analysis were implemented.

An interactive Power BI dashboard was developed to visualize key insights related to industry growth, fund performance, investor behavior, and SIP trends. The final solution provides investors and analysts with meaningful insights to support informed decision-making.

Problem Statement and Objectives

Problem Statement

The mutual fund industry generates large amounts of data related to fund performance, investor transactions, assets under management, and portfolio holdings. Investors often struggle to analyze this information effectively and identify suitable investment opportunities.

Traditional approaches rely on manual analysis and fragmented reports, making it difficult to evaluate risk, compare funds, and understand investor behavior.

Objectives

The primary objectives of this project are:

1. Build an automated ETL pipeline for mutual fund datasets.
2. Store cleaned data in a structured SQLite database.
3. Perform exploratory data analysis to identify trends and patterns.
4. Calculate important performance metrics.
5. Implement advanced risk analytics techniques.
6. Develop a mutual fund recommendation engine.
7. Create an interactive Power BI dashboard.
8. Generate actionable business insights.

Data Sources

The project uses multiple datasets representing different aspects of the mutual fund ecosystem.

Fund Master Dataset

Contains details about mutual fund schemes including:

- AMFI Code
- Scheme Name
- Fund House
- Category
- Plan Type

NAV History Dataset

Contains historical Net Asset Value records used for:

- Return calculations
- Risk calculations
- Performance comparisons

AUM Dataset

Contains assets under management information for different fund houses.

SIP Dataset

Contains monthly SIP inflow information used to analyze investor participation and industry growth.

Investor Transaction Dataset

Contains:

- Investor ID
- Transaction Type
- Transaction Date
- Investment Amount
- Demographic Information

These datasets collectively provide a complete view of the mutual fund ecosystem.

ETL Pipeline Design

Overview

ETL stands for Extract, Transform, and Load.

The ETL pipeline was developed to automate data preparation and ensure data quality before analysis.

Extraction Phase

Raw CSV files were collected from different sources and loaded into Python using Pandas.

Key Activities:

- Reading CSV files
- Validating schema
- Checking data completeness

Transformation Phase

Data cleaning operations included:

- Missing value handling
- Data type conversion
- Standardization
- Duplicate removal

These steps improved data consistency and reliability.

ETL Pipeline Design (Continued)

Loading Phase

After transformation, cleaned datasets were stored in:

- Processed CSV files

- SQLite Database

Benefits of ETL Pipeline

1. Improved data quality
2. Reduced manual effort
3. Faster analytics
4. Consistent reporting

Scripts Developed

- data_ingestion.py
- clean_transactions.py
- load_sqlite.py
- validate_amfi.py

The ETL process forms the foundation for all subsequent analytics activities.

Database Design

SQLite was used as the central storage layer for the project.

Main Tables

1. fund_master
2. nav_history
3. scheme_performance
4. investor_transactions
5. benchmark_indices

Relationships

AMFI Code was used as the primary linking field between tables.

Benefits of using SQLite:

- Lightweight
- Easy integration with Python

- Portable database file
- Fast querying for analytics

The database structure enabled efficient retrieval of information required for performance and risk analysis.

Page 8 - Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the datasets and identify meaningful patterns.

Areas Analyzed

1. Industry Growth
2. Fund Performance
3. Investor Behavior
4. SIP Trends
5. Portfolio Composition

Tools Used

- Pandas
- Matplotlib
- Jupyter Notebook

Key Questions Addressed

- How is the mutual fund industry growing?
- Which fund houses manage the largest assets?
- What investment patterns exist among investors?
- How have SIP inflows evolved over time?

EDA provided the foundation for deeper performance and risk analytics.

EDA Findings

Industry Analysis

Include:

- Industry AUM trend (2022- 2025)
- Monthly SIP inflow trend
- Folio growth trend

Findings:

- Mutual fund industry AUM showed consistent growth.
- SIP inflows increased steadily, indicating strong retail participation.
- Folio counts increased, reflecting growing investor awareness.

Investor Analysis

Include:

- State-wise investment chart
- Age-group distribution
- Transaction type distribution

Findings:

- Metropolitan cities contributed the highest investment volume.
- SIP transactions formed the majority of investments.
- Investors aged 26- 45 represented the largest investor segment.

Performance Analytics (2 Pages)

Metrics Calculated

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Results

cagr.csv

max_drawdown.csv

alpha_beta.csv

sharpe_sortino.csv

A	B	C	D
amfi_code	cagr		
100016	0.026352		
100025	0.044551		
100033	0.300997		
101206	0.235205		
101207	0.079331		
101208	0.065044		
102885	0.182188		
102886	0.011709		
102887	0.168287		
118632	0.240312		
118633	0.162622		
118634	0.163614		
118635	0.159455		
118636	0.053132		
119092	0.063635		
119093	0.078919		
119094	0.281926		
119095	0.015205		
119120	0.058853		
119551	0.257849		
119552	0.21508		
119598	0.323981		
119599	0.020531		
120503	0.187713		
120504	0.232774		
120505	0.328016		
120506	0.170722		
120507	0.072324		
120841	0.135223		
120842	0.066968		
120843	0.308833		
120844	0.069104		
125497	0.190036		
125498	0.112937		

amfi_code	max_drawdown
100016	-0.24734
100025	-0.04308
100033	-0.16217
101206	-0.11292
101207	-0.35447
101208	-0.00162
102885	-0.1086
102886	-0.28001
102887	-0.2154
118632	-0.17414
118633	-0.1863
118634	-0.23345
118635	-0.11651
118636	-0.08316
119092	-0.14402
119093	-0.21751
119094	-0.20961
119095	-0.51678
119120	-0.04329
119551	-0.15012
119552	-0.11803
119598	-0.28706
119599	-0.52574
120503	-0.13898
120504	-0.10500

amfi_code	alpha	beta
100016	0.037476	-0.05827
100025	0.042818	0.001158
100033	0.271954	0.005104
101206	0.213998	0.021086
101207	0.108971	-0.06529
101208	0.060861	0.000267
102885	0.170488	-0.01949
102886	0.028969	-0.04213
102887	0.162113	0.016683
118632	0.218294	-0.00835
118633	0.156438	-0.03664
118634	0.175007	0.103497
118635	0.151364	-0.00143
118636	0.050748	0.001257
119092	0.068995	0.009731
119093	0.082328	0.025883
119094	0.260767	-0.06627
119095	0.048016	-0.06695
119120	0.056209	-0.00641
119551	0.23201	-0.03175
119552	0.198686	-0.02616
119598	0.30337	-0.0232
119599	0.048824	0.062002
120503	0.177033	-0.04027
120504	0.211948	0.016232

amfi_code	sharpe_ratio	sortino_ratio
100016	-0.20152	-0.35105
100025	-0.56709	-0.94182
100033	1.093699	1.829134
101206	1.027213	1.799563
101207	0.162661	0.276644
101208	-0.81557	-1.68104
102885	0.817099	1.435742
102886	-0.20583	-0.34722
102887	0.619518	1.088755
118632	1.081659	1.850133
118633	0.645207	1.154476
118634	0.448434	0.746015
118635	0.664857	1.137977
118636	-0.35663	-0.61353
119092	0.030785	0.052574
119093	0.129614	0.221087
119094	0.998231	1.703797
119095	-0.07597	-0.12819
119120	-0.22658	-0.37677
119551	1.208267	2.140267
119552	0.953279	1.609776
119598	0.945308	1.675317
119599	-0.05719	-0.09439
120503	0.797973	1.337943
120504	1.026524	1.805294

Findings

- **Several funds outperformed benchmark returns.**
- **Higher Sharpe Ratio funds delivered better risk-adjusted returns.**
- **Low drawdown funds showed better downside protection.**

Advanced Analytics

Historical VaR & CVaR

Purpose:

- **Measure downside risk.**

Include:

var_cvar_report.csv

amfi_code	VaR_95	CVaR_95	
100016	-0.01436	-0.01806	
100025	-0.00379	-0.00499	
100033	-0.01903	-0.02346	
101206	-0.01328	-0.01744	
101207	-0.02602	-0.03246	
101208	-0.00027	-0.00042	
102885	-0.01261	-0.01549	
102886	-0.01922	-0.02325	
102887	-0.01523	-0.01941	
118632	-0.01395	-0.01762	
118633	-0.01344	-0.01663	
118634	-0.02544	-0.0323	
118635	-0.01255	-0.01618	
118636	-0.0038	-0.00492	
119092	-0.01375	-0.01733	
119093	-0.01423	-0.01749	
119094	-0.01848	-0.02426	
119095	-0.02619	-0.03167	
119120	-0.00394	-0.00501	
119551	-0.01285	-0.0164	
119552	-0.0135	-0.01734	
119598	-0.02451	-0.0306	
119599	-0.02686	-0.03238	
120503	-0.01355	-0.01745	
120504	-0.01373	-0.01749	

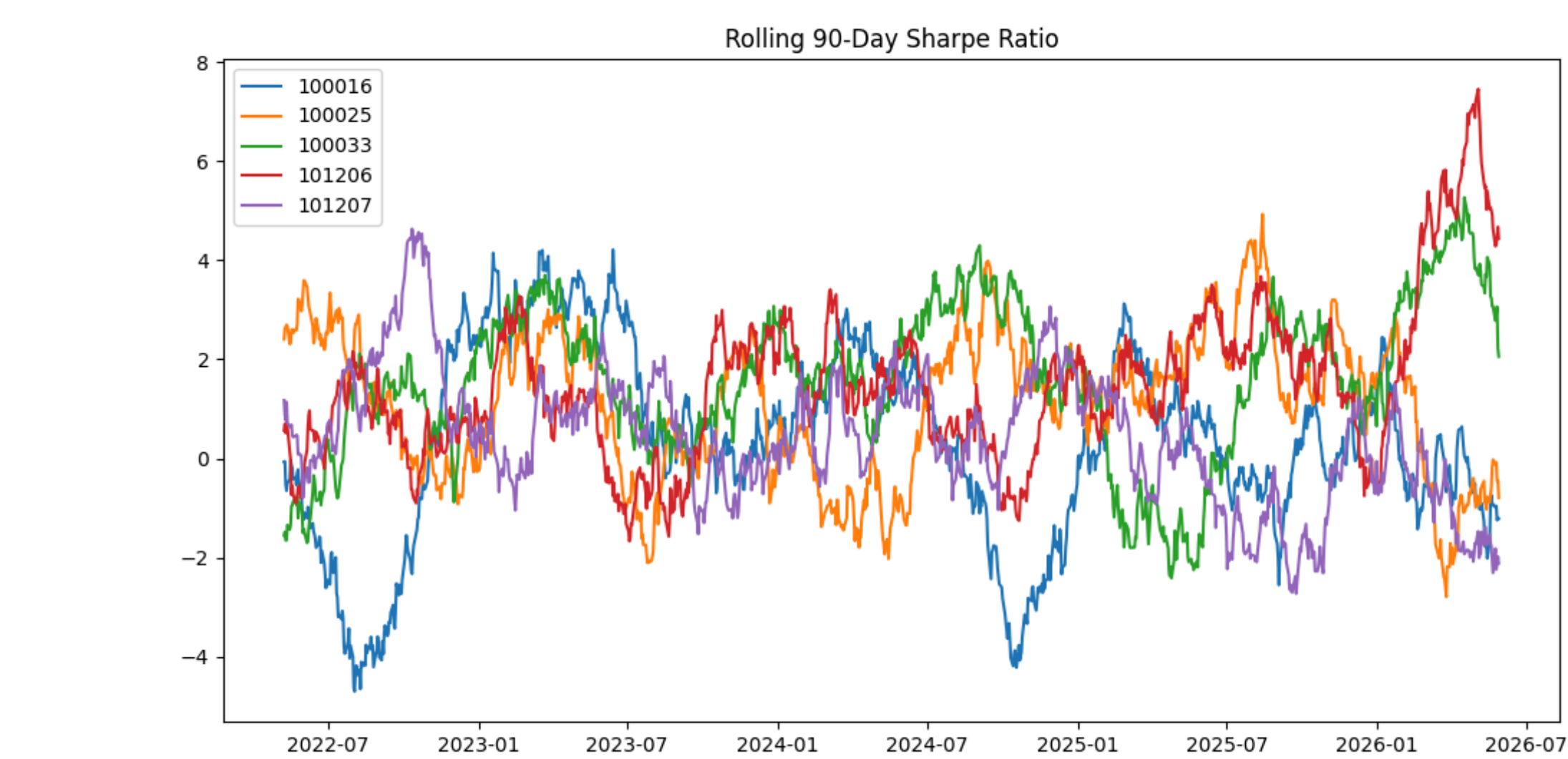
<

>

var_cvar_report

Rolling Sharpe Ratio

Include:



Investor Cohort Analysis

Include:

cohort_year	avg_sip_amt	total_invested	investor_count
2024	107739.5	2.5E+09	4803
2025	106704.3	1.02E+09	4018

SIP Continuity Analysis

investor_id	sip_count	avg_gap_days	status
INV000004	6	85.4	At Risk
INV000008	6	70.4	At Risk
INV000010	6	64.8	At Risk
INV000011	7	40.16667	At Risk
INV000012	8	57	At Risk
INV000013	7	55.33333	At Risk
INV000014	7	75.33333	At Risk
INV000023	8	58.57143	At Risk
INV000028	6	93.6	At Risk
INV000029	7	60.66667	At Risk
INV000030	9	47.625	At Risk
INV000033	8	45.57143	At Risk
INV000039	6	68.6	At Risk
INV000045	6	70.6	At Risk
INV000048	9	46	At Risk
INV000052	6	36.2	At Risk
INV000056	6	74.8	At Risk
INV000057	8	49.71429	At Risk
INV000069	10	49.11111	At Risk
INV000072	6	83	At Risk
INV000073	9	48.25	At Risk
INV000074	8	72.57143	At Risk
INV000079	6	74.8	At Risk
INV000076	6	91.8	At Risk
INV000079	6	65.8	At Risk

Fund Recommendation System

Example output:

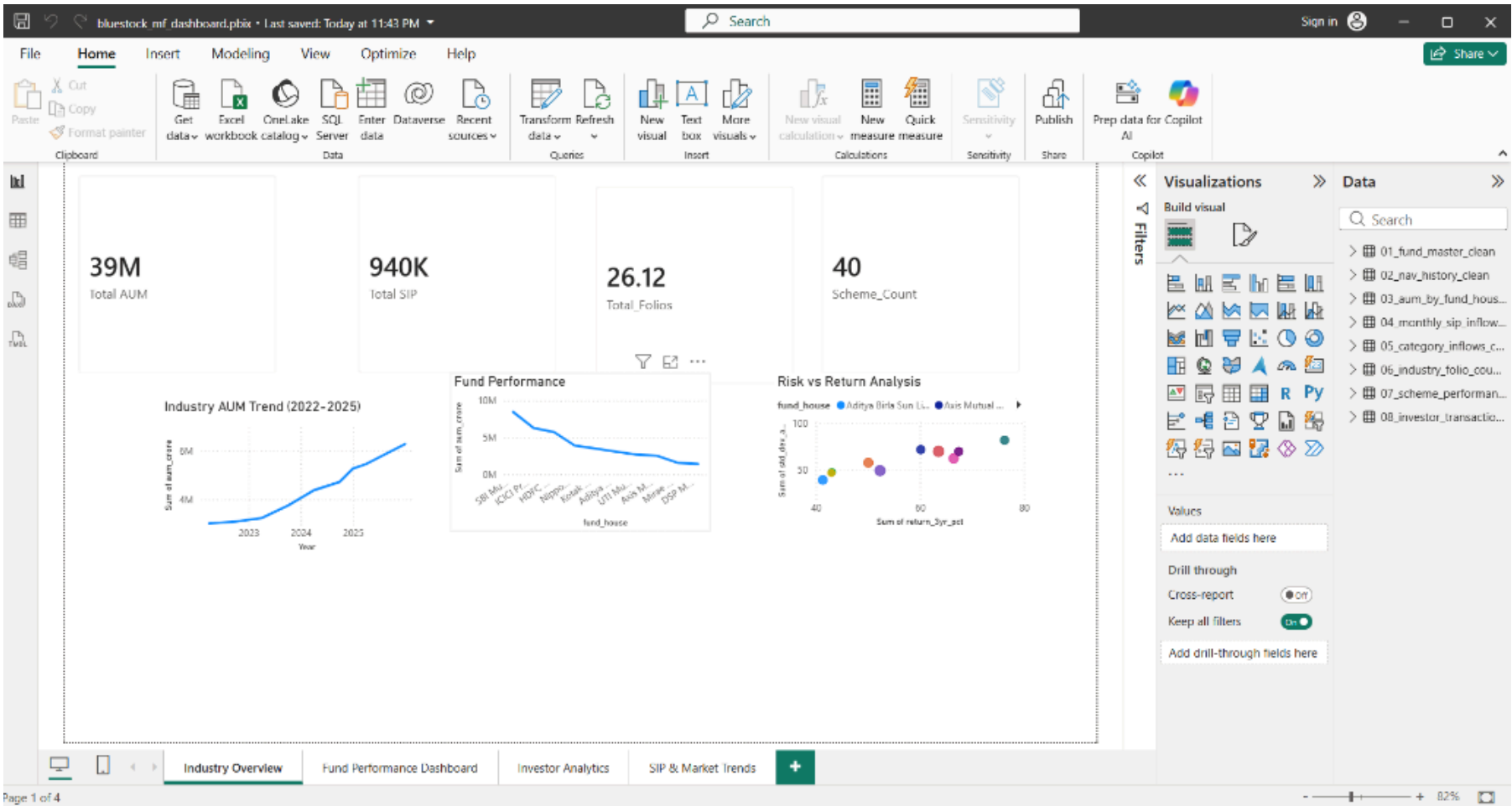
HDFC Top 100 Fund
Mirae Asset Large Cap Fund
ICICI Prudential Bluechip Fund
HHI Concentration Analysis

Explain:

- High HHI → concentrated portfolio
- Low HHI → diversified portfolio

Dashboard Analysis

Industry Overview



Discuss:

Total AUM

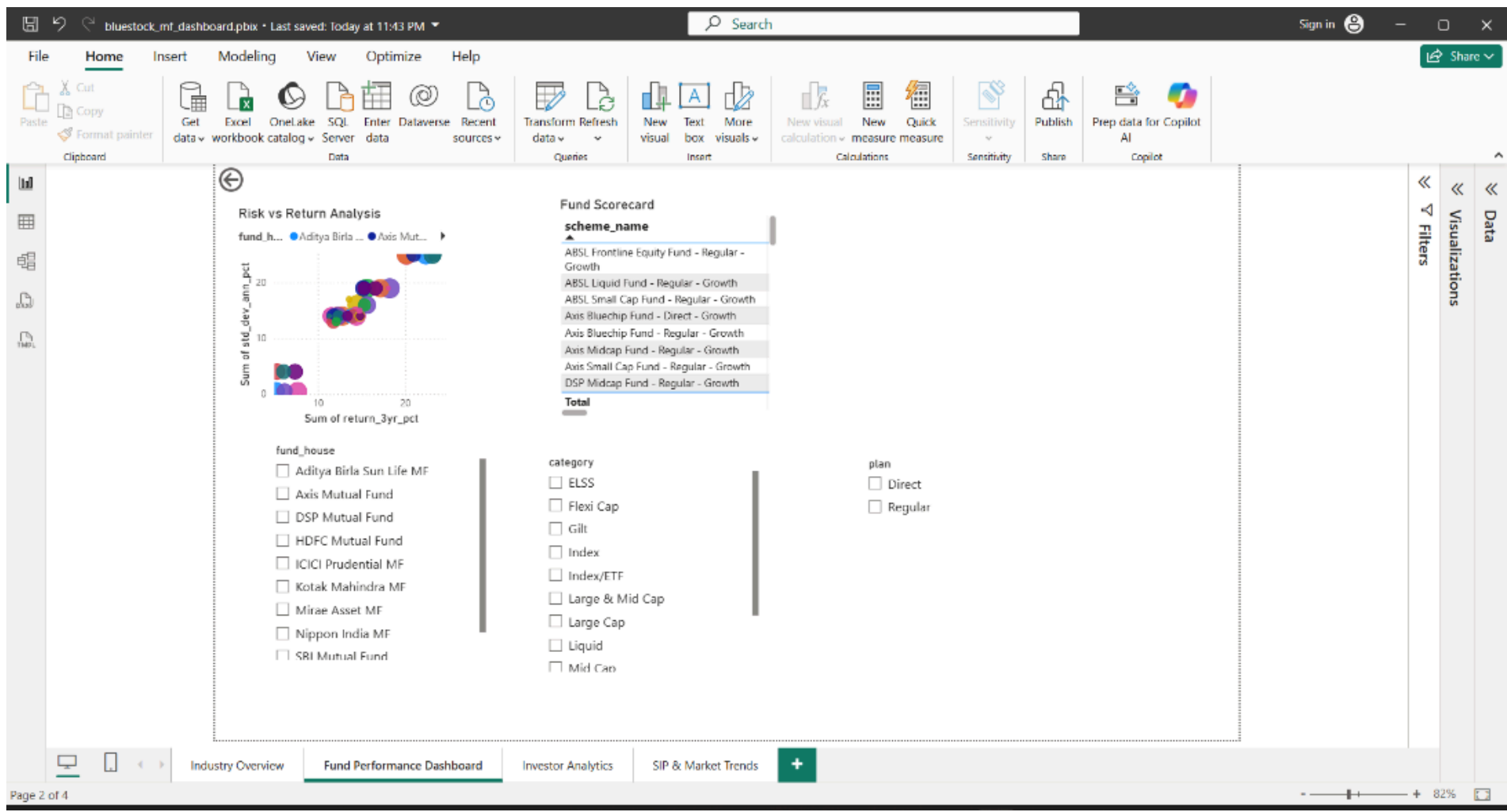
Total SIP Inflows

Total Folios

Total Scheme

Fund Performance

page2_fund_performance.png

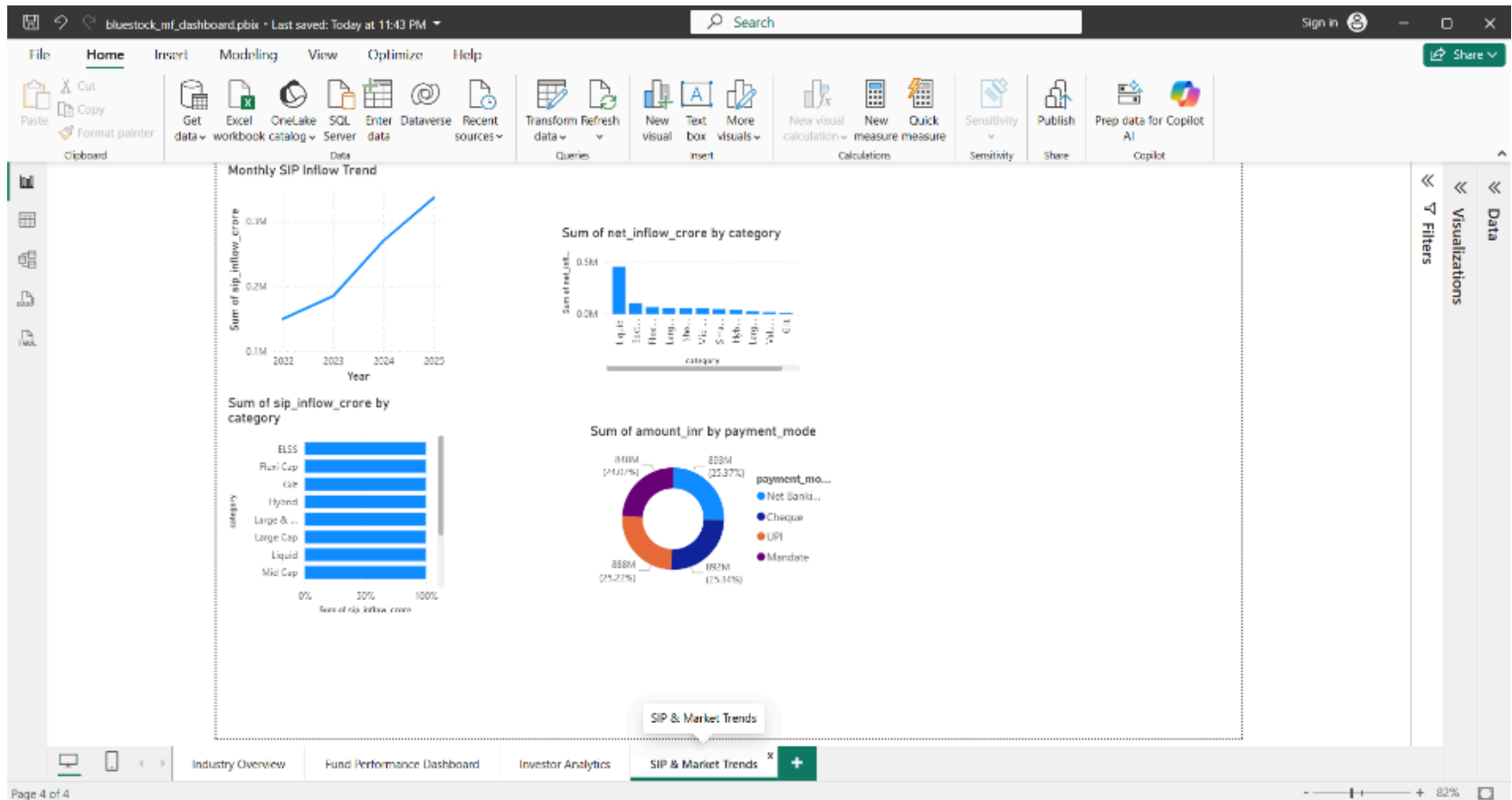
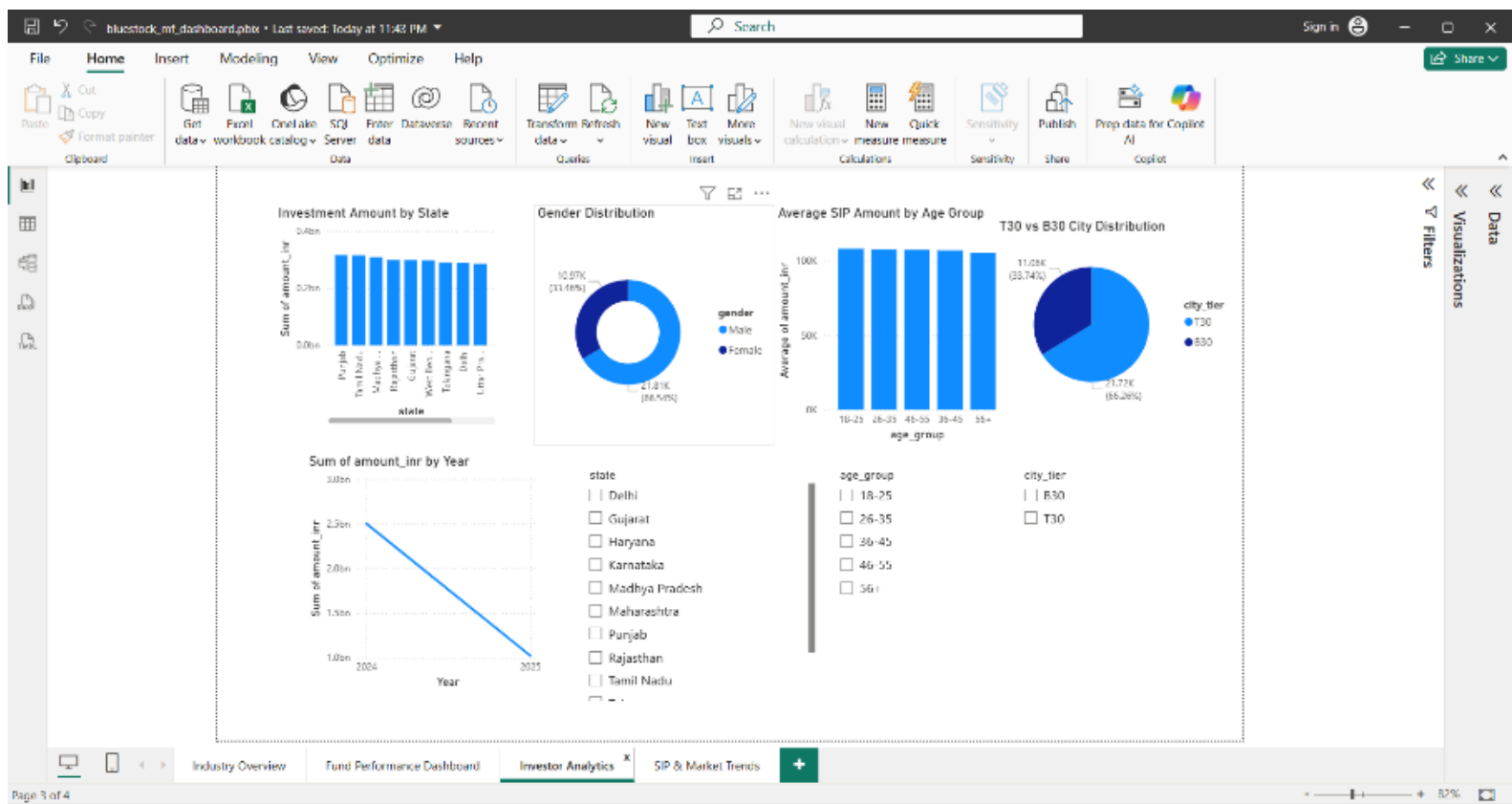


Discuss:

- Return vs Risk Scatter Plot
- Fund Scorecard
- NAV vs Benchmark Comparison

Investor Analytics & SIP Trends

page3_investor_analytics.png , page4_sip_market_trends.png



Key Findings

1. Mutual Fund Industry AUM Demonstrated Strong Growth

- Assets Under Management (AUM) increased consistently during the analysis period.
- Growth indicates rising investor confidence in mutual fund investments.
- Leading fund houses contributed significantly to overall industry expansion.
- Increasing AUM reflects the growing popularity of professionally managed investment products.

2. SIP Contributions Remained the Primary Driver of Industry Expansion

- Monthly SIP inflows showed a steady upward trend.
- Investors increasingly preferred systematic investing over lump-sum investments.
- SIPs promoted disciplined investing and long-term wealth creation.
- Rising SIP contributions indicate improved financial awareness among retail investors.

3. Moderate-Risk Funds Generated Strong Risk-Adjusted Returns

- Several moderate-risk funds achieved high Sharpe Ratios.
- These funds provided a balanced combination of return and risk.
- Investors received competitive returns without excessive volatility.
- Moderate-risk funds emerged as suitable options for long-term investors.

4. Investor Participation Was Highest Among Working-Age Groups

- Investors aged 26- 45 accounted for the largest share of investments.
- This group showed higher participation in SIP-based investing.
- Stable income levels and long-term financial goals contributed to higher investments.
- Working professionals demonstrated strong interest in wealth accumulation through mutual funds.

5. Diversified Portfolios Exhibited Lower Concentration Risk

- **HHI analysis showed diversified portfolios had lower risk exposure.**
- **Funds with broader holdings were less affected by sector-specific downturns.**
- **Diversification improved portfolio stability during market fluctuations.**
- **Well-diversified funds offered better risk management and long-term sustainability.**

Recommendations

1. Encourage Long-Term SIP Investments

- **Promote SIPs as a disciplined and low-risk investment approach.**
- **Educate investors about the benefits of rupee-cost averaging.**
- **Encourage long-term investment horizons to maximize wealth creation.**
- **Increase awareness through investor education programs and campaigns.**

2. Promote Diversified Portfolios

- **Encourage investors to spread investments across multiple sectors and asset classes.**
- **Reduce dependence on a small number of securities.**
- **Improve portfolio stability during market volatility.**
- **Regularly review portfolio allocations to maintain diversification.**

3. Monitor Funds with High VaR and Maximum Drawdown

- **Investors should evaluate downside risk before investing.**
- **Fund managers should closely monitor schemes with high VaR values.**
- **Risk mitigation strategies should be implemented for highly volatile funds.**
- **Regular risk assessments can help improve investor confidence.**

4. Improve Investor Retention Through SIP Continuity Monitoring

- **Identify investors with irregular SIP contributions.**

- **Send reminders and notifications for missed SIP payments.**
- **Develop engagement programs for at-risk investors.**
- **Monitor SIP discontinuation rates and take corrective actions.**

5. Enhance the Recommendation System Using Machine Learning

- **Incorporate investor profiles, goals, and historical preferences.**
- **Use predictive analytics to improve recommendation accuracy.**
- **Introduce personalized investment suggestions.**
- **Continuously update models using new market and investor data.**

6. Expand Mutual Fund Awareness in Underrepresented Regions

- **Conduct financial literacy programs in smaller cities and towns.**
- **Increase awareness about mutual fund benefits and investment options.**
- **Partner with local financial institutions to improve outreach.**
- **Promote digital investment platforms to improve accessibility.**

References:

AMFI India

NSE India

Power BI Documentation

Python Documentation

Pandas Documentation

NumPy Documentation

Matplotlib Documentation

SQLite Documentation

Thank You